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GUIDELINE FOR TRANSPORTATION SECURITY FOR RADIOACTIVE MATERIAL

This guideline specifies the basic security conditions for obtaining a transport authority for Group IV hazardous substances as defined in the Hazardous Substances Act 15 of 1973, referred to in this guideline as radioactive material. This represents the guidance of the South African Health Products Regulatory Authority (SAHPRA) applicable to all governmental and private organisations involved in the transportation of Group IV hazardous substances. In the event of conflict with any transport security-related regulations, such regulations will take precedence.

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2		

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1. Preliminaries

1.1 Purpose

This guideline specifies the basic security conditions for obtaining a transport authority for Group IV hazardous substances as defined in the Hazardous Substances Act 15 of 1973, referred to in this guidance as radioactive material. This guidance is provided in support of the Regulations relating to Group IV Hazardous Substances, issued in terms of section 29 of the Hazardous Substances Act (15 of 1973) and published under Government Notice R247 in Government Gazette 14596, dated 26 February 1993 (R247) and any subsequent regulations made pursuant to the Hazardous Substances Act.

This guidance is not intended to relieve an authorised party of the duty to take any additional actions as may be appropriate and necessary for radiological safety and security to protect the public, patients, radiation workers and the environment.

1.2 Scope

This guidance applies to all the transport and associated activities of all Category 1, 2, and 3 radioactive materials. This guidance does not apply to nuclear material (as defined in the Nuclear Energy Act, No. 46 of 1999), except for radioactive sources that incorporate plutonium-239.

This guidance applies to road, air, rail, and maritime transport of radioactive material both domestically and internationally and must be adhered to by all conveyors, users, and manufacturers.

1.3 Fundamental Obligation

Any person making an application for the transport of radioactive material must comply with this guidance and the security-related requirements and conditions of the applicable authority as a condition for receiving authorisation.

2. Definitions

Definitions included in the Hazardous Substances Act 15 of 1973 shall also apply to this guideline. **For the purpose of this guideline, the following additional definitions apply:**

Act	The Hazardous Substances Act 15 of 1973
Associated activity	The acquisition, disposal, possession, production, use, storage, import, export or transport of radioactive material.
Associated facility	A radioactive material facility
Authority Holder	Any person, organisation, or government entity authorised by the Regulator to engage in an associated activity or operate an associated facility. In relation to the transport of radioactive material, Authority Holders include the Consignor, Conveyor and Consignee.
Authority	A written authority referred to in section 3A of the Act
Consignee	Any person or organisation that is authorised to take delivery of a consignment of radioactive material (authorised recipient)
Consignor/shipper	Any person or organisation that prepares or offers a consignment of radioactive material for transport (authorised sender)
Contingency Plan	Predefined sets of actions for response to unauthorised acts indicative of attempted unauthorised removal or sabotage, including threats, designed to effectively counter such acts
Conveyor/Carrier	Any person or organisation that is authorised to transport/convey radioactive material
Delay	The act of postponing hindering or causing something to occur more slowly
Detection	A process in a security system that begins with sensing a potentially malicious or other unauthorised act and that is completed with the assessment of the cause of the alarm
Export	The physical transfer of radioactive material originating from the Republic of South Africa to an importing State or to a recipient in an importing State
Graded approach	The application of radiological security measures proportional to the potential consequences of a malicious act
Group IV Hazardous Substance	means radioactive material which is outside a nuclear installation as defined in the Nuclear Energy Act, 1999, and is not a material which forms part of or is used or intended to be used in the nuclear fuel cycle, and - (a) has an activity concentration of more than 100 becquerels per gram and a total activity of more than 4 000 becquerels; or (b) has an activity concentration of 100 becquerels or less per gram or a total activity of 4 000 becquerels or less and which the Minister has by notice, in the <i>Gazette</i> declared to be a Group IV hazardous substance, and which is used or intended to be used for medical,

	scientific, agricultural, commercial or industrial purposes, and any radioactive waste arising from such radioactive material; <i>[Definition of “Group IV hazardous substance” inserted by s. 1 of Act 53/92, substituted by s. 84 of Act 131/93 and amended by s. 58 of Act 46/99]</i>
Guard	A person who is entrusted with responsibility for patrolling, monitoring, assessing, escorting individuals or transport, controlling access and or providing initial response
IAEA	International Atomic Energy Agency
Import	The physical transfer of radioactive material originating from an exporting State into the Republic of South Africa or to a recipient in South Africa
Malicious Act	An act or attempted act of sabotage or of unauthorised removal of radioactive material
Management	Operating an associated facility or engaging in an associated activity, including transport
NECSA	South African Nuclear Energy Corporation
Performance testing	Testing of security measures and the security system to determine whether or not they are implemented as designed; adequate for the proposed natural, industrial and threat environments; and in compliance with established performance requirements
R247	The Regulations relating to Group IV Hazardous Substances, made in terms of section 29 of the Hazardous Substances Act 15 of 1973 and published under Government Notice R247 in Government Gazette 14596, dated 26 February 1993 (R247)
Radioactive material	Any Group IV hazardous substance as defined in the Hazardous Substances Act 15 of 1973
Radiological security event	An event that is assessed as having implications for radiological security.
Radiological Security Officer	A person appointed as such in terms of Regulation 13 (3), including an Acting Radiological Security Officer
Radiological security	The prevention and detection of, and response to, theft, sabotage, unauthorised access, illegal transfer or other malicious acts involving radioactive material, associated activities or associated facilities
Regulatory authority/ Regulator	South African Health Products Regulatory Authority
Response	The actions undertaken, following detection, to prevent an adversary from succeeding in unauthorised removal or sabotage or to mitigate potentially harmful consequences.
Sabotage	Any deliberate act directed against an associated facility or an associated activity that could directly or indirectly endanger the health and safety of the public, patients, radiation workers or the environment by exposure to radiation or release of radioactive substances.
Safety	Measures intended to minimise the likelihood of accidents involving radioactive material and, should such an accident occur, mitigate its consequences.

Special form	Special form certificate is an official document that certify that a radioactive source meets international safety standards under IAEA and UN regulations. A sealed source that is solid, encapsulated and leak tight, such that it cannot disperse under normal transport conditions- this ensure that even if the source is damaged or dropped, it will not leak radioactive material.
Security culture	The assembly of characteristics, attitudes and behaviours of individuals, organisations and institutions which serve to support, enhance and sustain radiological security
Security management	The establishment and implementation of policies, plans, and procedures, and the deployment of the necessary resources, for the security of radioactive material associated facilities and associated activities. Security management includes measures for access control, trustworthiness verification, information protection, computer security, preparation of a Security Plan, training and qualification of personnel, accounting, inventory and response to and reporting of nuclear security events.
Security Plan	A document prepared by the Authority Holder (or applicant for an authority) that presents a detailed description of the security arrangements in place at an associated facility or in connection with an associated activity.
Storage in Transit	The holding of radioactive material in a location incidental to transport that provides security commensurate with the category of material being transported.
Threat	A person or group of persons with motivation, intention and capability to commit a malicious act.
Trustworthiness determination	An assessment of an individual's integrity, honesty and reliability; pre-employment checks and checks during employment that are intended to identify the motivation or behaviour of persons with authorised access to associated facilities or associated activities or to sensitive information.
Unauthorised removal	The theft or other unlawful taking of radioactive material.

3. Responsibilities

3.1 Responsibilities of the Authority Holder

The person holding the Authority to Convey is responsible for security during transport.

3.2 Qualification and Training

The Authority Holder shall ensure that all personnel on whom security depends are appropriately trained and qualified so that they understand their responsibilities and can perform their duties with appropriate

judgement and according to defined procedures. They shall be periodically retrained or re-qualified as may be appropriate.

At a minimum, such training shall include:

- (1) All employees shall be informed at least annually of the importance of effective security measures and be trained in their implementation as appropriate.
- (2) Records shall be kept of such training and information sessions. The records shall include at a minimum the date, details of presenter(s), details of content, and details of attendees.
- (3) Training programmes shall be routinely evaluated and updated, as necessary.

Individuals engaged in the transport of radioactive material should receive security training, including Basic Security Awareness Training. This training should include training on the need for transport security, the nature of security-related threats, methods to address security concerns and actions to be undertaken if a radioactive material security event occurs. It should also include awareness of Transport Security Plans when appropriate, commensurate with the responsibilities of individuals and their roles in implementing Transport Security Plans.

3.3 Transfer of Radioactive Material

The Authority Holder shall not transfer (hand over) radioactive material to another party unless:

- (1) the Authority Holder is authorised to do so by SAHPRA and
- (2) the recipient possesses a valid authority for the radioactive material; and
- (3) the Authority Holder provides the recipient with all relevant technical information to permit the safe and secure management of the radioactive material.

3.4 Export of Radioactive Material

Any person intending to export radioactive material shall apply to the Regulator for an export authority.

- (1) The application for authority to export radioactive material shall include a copy of the recipient's authorisation to receive and possess the radioactive material to be exported. **The recipient's authorisation shall include at least the following information:**
 - (a) Name of the issuing authority;
 - (b) Name of the recipient;

- (c) Recipient location and legal address or principal place of business;
- (d) Name and contact details (phone number and email address) of responsible person;
- (e) Relevant radionuclides;
- (f) Activity level of the radionuclides (TBq);
- (g) Uses of the radioactive material;
- (h) Container to be used and a valid container certificate;
- (i) Manufacturer source certificate; and
- (j) Recipient authorisation expiry date (if any).

(2) Other information to be submitted as part of the application for authority to export may include, as applicable:

- (a) Chemical and physical form (e.g., solid, liquid, gas, capsule, resin);
- (b) Reason for export, e.g., sale, contract work, or returning depleted sources to manufacturer;
- (c) Type of packaging according to the IAEA *Regulations for the Safe Transport of Radioactive Material*;
- (d) Justification or explanation of any need to use the “exceptional circumstances” provisions in IAEA ‘Guidance on the Import and Export of Radioactive Sources’, if applicable;
- (e) Modes of transport (air, sea, road or rail) and name and Authority number of the Conveyor;
- (f) Port of exit and estimated date of export. *(This information needs to be protected. The transmission of the dates may be in a separate enclosure or via password protected file. Access to this information must be for need-to-know personnel only, both with the exporter and SAHPRA.)*

(3) After receiving an authority to export Category 1 or 2 radioactive material, the Authority Holder shall ensure that:

- (a) The export of the radioactive material is conducted in compliance with all applicable transport requirements of the IAEA *Regulations for the Safe Transport of Radioactive Material* and other, national Regulations on transport of dangerous goods, the IAEA Code

of Conduct and its 'Supplementary Guidance on the Import and Export of Radioactive Sources'; and

- (b) The regulatory body or other relevant counterpart in the importing State is notified through the Regulator in advance (at least seven days to the extent practicable) of each shipment with the following information in writing:
 - i. the estimated date of export (This information needs to be transmitted securely as well).
 - ii. exporting Authority Holder
 - iii. recipient's name and Import authority (license) number, and contact details of responsible person
 - iv. radionuclide(s)
 - v. Activity level of the radionuclides (TBq);
 - vi. the number of radioactive sources and, if available, their unique identifiers.

3.5 Confirmation of Export

The Export Authority Holder shall provide the Regulator with documented evidence that all the radioactive material listed on the export authority have been cleared by the South African Revenue Services (SARS) Customs and the exportation has been concluded, or that the export transaction has been cancelled. This notification must occur within one month of completion of the exportation or cancellation.

3.6 Import of Radioactive Material

An Authority Holder intending to import radioactive material shall apply to SAHPRA for an import authority.

- (1) **The application for authority to import radioactive material shall include the following information:**
 - (a) Exporting country;
 - (b) Name and email address of the regulatory body in the exporting state;
 - (c) Name, address and contact details (phone number and email address) of the exporter;
 - (d) Name and contact details (phone number and email address) of responsible person representing the recipient;

- (e) Recipient location and legal address or principal place of business;
 - (f) File number and authority number of end user's authority to possess and use the radioactive material;
 - (g) For each source: radionuclide name;
 - (h) Activity level of the radionuclides (TBq);
 - (i) Security category of the consignment;
 - (j) Intended use of the radioactive material, whether sealed or unsealed, and physical form (e.g., solid, liquid, gas, capsule, resin);
 - (k) Type of packaging according to the IAEA *Regulations for the Safe Transport of Radioactive Material*;
 - (l) Details of the arrangements for the safe and secure disposal or management of the source(s), including financial provisions where appropriate, once they have become disused, including copies of any contractual agreements;
 - (m) Justification or explanation of any need to use the "exceptional circumstances" provisions in IAEA 'Guidance on the Import and Export of Radioactive Sources', if applicable;
 - (n) Modes of transport (air, sea, road or rail) and name and file number of end conveyor; and
 - (o) Port of entry and expected date of entry.
- (2) After receiving authority to import the radioactive material, the Authority Holder shall, to the extent possible, ensure that the import of the radioactive material complies with all applicable transport requirements of the IAEA Regulations for the Safe Transport of Radioactive Material and with other national Regulations for transport of dangerous goods.

3.7 Confirmation of Import

The Authority Holder shall provide the Regulator with documented evidence that all the radioactive materials listed on the import authority have been cleared by SARS Customs and the importation has been concluded, or that the import transaction has been cancelled. This notification must occur within one month of completion of the importation or cancellation.

3.8 Reporting of Radiological Security Events

The authority holder shall include provisions in the Transport Security Plan to report and take action if a situation involving the loss of control of unauthorised access to actual or attempted theft or sabotage, or misrouting during transport of radioactive material occurs. **The Authority Holder shall include provisions in the Transport Security Plan for reporting and taking action as follows:**

- (1) Immediately inform SAHPRA and SAPS Explosives Unit;
- (2) Take appropriate action to remedy the circumstances and to prevent a recurrence of similar situations;
- (3) Investigate the causes, circumstances, and consequences of the event and
- (4) Within seven days, or as required, provide the Regulator with a report on the causes of the event, its circumstances and consequences, and on the corrective or preventive actions taken or to be taken.

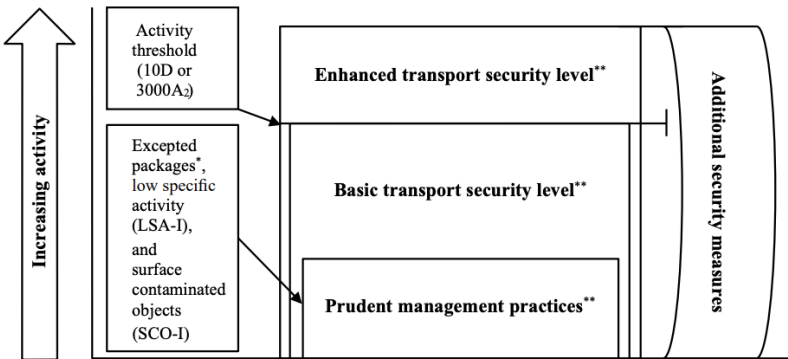
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4. Categorisation and Assignment of Security Level

Radioactive materials shall be assigned a security category of Enhanced Transport Security Level or Basic Transport Security Level. Packages containing Category 1 or 2 radioactive sources are assigned to the Enhanced Transport Security Level, and packages containing Category 3 radioactive sources are assigned to the Basic Transport Security Level. **The category of the material is assigned based on the activity of the sources using the methodology described below:**

- The D-value is a radionuclide-specific value assigned to radioactive material which, if not under control, could cause severe deterministic effects for a range of scenarios that include both external exposure from unshielded radioactive material and inadvertent internal exposure following dispersal (e.g., by fire or explosion) of the radioactive material.
- The activity of radioactive material (A) in common applications varies over many orders of magnitude. D-values are therefore used to normalise the range of activities to provide a reference in comparing risks. This is done by taking the activity A of the source (in TBq) and dividing it by the D-value for the relevant radionuclide.
- Authority Holders shall categorise their radioactive material in accordance with Annex I of this document. Authority Holders shall use the A/D ratio as shown in Table 1 for categorising radioactive sources with an assigned D value as listed in Table 2.
- For unsealed material, if the activity is greater than $3000A_2$, it falls within the Enhanced Transport Security Level. If below $3000A_2$, it falls within the Basic Transport Security Level. A_2 values can be found in Table 2 of IAEA Publication Regulations for the Safe Transport of Radioactive Material (No. SSR-6 Rev. 1)

See Figure 1 below for a graphic representation of the security levels.



*Excepted packages with activity not exceeding the level permitted for the radioactive material when it is not in special form (see para. 422 of Ref. [14]).

**The regulatory body may require additional security measures in addition to any security measures already prescribed for a given transport security level.

FIGURE 1 TRANSPORT SECURITY LEVELS (NSS 9-G)

5. Requirements for Security in the Transport of Radioactive Material

5.1 Prudent Management Practices

Prudent Management Practices should be applied to all shipments of radioactive material regardless of material category. These practices are the same basic security precautions that might be applied to any cargo of high value.

1) General security provisions

- a) The Authority Holder shall ensure that all persons engaged in the transport of radioactive material apply security measures commensurate with their responsibilities and the level of threat.
- b) Radioactive material shall be transferred only to authorised Authority Holders.
- c) When radioactive material is temporarily stored at transit sites or makes an expected or unexpected stop, appropriate security measures shall be applied consistent with the measures applied during use and storage.
- d) The Authority Holder shall have procedures in place to ensure that, if any package is not delivered to the intended recipient at the expected time, an inquiry is initiated about the status of that package. If the package has been lost or stolen or appears to have been tampered with, procedures shall immediately be initiated to locate and recover the package.
- e) Enclosed lockable vehicles should be used for transportation of radioactive materials.
- f) Radioactive material must be transported in approved packaging and the transport package must be locked and secured to the vehicle to prevent inadvertent movement during transport.
- g) Transport containers must be locked inside the vehicle and must be positioned at the maximum distance possible from the passenger compartment.
- h) If an open vehicle is used, the transport container must be permanently fixed to the vehicle unless it weighs more than 2000 kg.
- i) Utilising carriers with package tracking systems (e.g., bar code system to monitor the status of the shipment), as appropriate;
- j) Not leaving packages or conveyances unattended for any longer than is absolutely necessary;

- k) Providing drivers of road conveyances with effective communication capability;
- l) Satellite tracking system on vehicle.

5.2 Basic Transport Security Level Requirements (Categories 1, 2, and 3)

In addition to Prudent Management Practices, the Basic Transport Security Level Requirements shall be applied to radioactive material that has been evaluated as Category 1, 2, or 3.

5.2.1 Contingency Planning

Contingency plans include predefined sets of actions for response to off-normal situations, such as unauthorised acts indicative of attempted unauthorised removal or sabotage, including threats thereof designed to effectively counter such acts.

Contingency plans must be developed for all shipments of Category 1, 2, and 3 radioactive materials. The graded approach will be applied such that the level of rigor and detail will be greater for higher category material, with Category 1 being the highest category and requiring the most detail.

Category 3 Contingency Plans should include emergency contact telephone numbers and basic driver actions to take if an off-normal event occurs.

Category 2 Contingency Plans should include emergency contact telephone numbers of detailed instructions on actions to take if an off-normal event occurs.

Category 1 Contingency Plans should include comprehensive instructions for response to off-normal situations based on an analysis of all credible hazards and threats which may arise during the shipment.

General notes on developing contingency plans:

- All safety incidents have a security component if the shipment is halted.
- Contingency plans are a working document containing formats, such as flow-charts and bulleted lists.
- The credible hazards and threats may be identified using a standardised checklist.
- Identify categories of risks using the checklist, for example:
 - o Environmental/Natural
 - o Traffic/Routing
 - o Medical
 - o Mechanical

- o Malicious act
- Each risk category should have a response framework developed so that an individual contingency plan does not need to be developed for every conceivable scenario – the basic actions are the same within each category. If there is a major difference in protocol, then a new plan can be developed (i.e., if there is severe weather inbound, vs a mudslide that damages the road – the severe weather might cause a delay in the shipment, while the mudslide might reroute the shipment, but both natural disasters).
- Contingency plans must always support and not hinder first responders, and should never dictate first responder actions (i.e., do not tell the police what to do, and need to be compatible with the police response so that they do not slow down or reduce the effectiveness of the police response).
- It is acceptable for security and safety contingency plans to be combined in a single document so that the driver has access to all contingency plans in a single location.

5.2.2 Trustworthiness Determination

The Authority Holder shall subject persons engaged in the transport of radioactive material to a trustworthiness determination commensurate with their responsibilities.

Methods of determining trustworthiness:

- 1) Criminal history check conducted by the South African Police Service (SAPS).
- 2) Conflict of interest statement, examples might include self-reported disclosures of any outside business dealings, which could cause a violation of this security guidance or incentives to compromise the security of radioactive material in transport. This statement should be reviewed and approved by a senior management official of the Authority Holder.
- 3) Drug and Alcohol testing should be administered at randomised intervals and should ensure that personnel performing their duties in the transport of radioactive material are not performing their duties while impaired.

5.2.3 Shipper and Carrier Credentials

Each crew member of any conveyance transporting radioactive material shall carry means of identification, such as an official identification document, including an identity document, driver's license, or passport.

This photo identification must be accompanied by a written authorisation from the crew member's employer, with the names of individuals specifically authorized to perform a given transport.

5.2.4 Written Procedures

The Authority Holder shall develop implementation procedures and provide them to crew members. These procedures shall address the application of security measures which have been required as a condition of the Transport Authority issued by the regulatory body.

Response procedures should include basic steps on how to respond to an off-normal incident during transport, such as safe parking instructions and emergency points of contact. At a minimum, these procedures should include 24-hour contact information for the Authority Holder's radiation protection officer, the regulatory body, and SAPS. Additionally, they may include emergency contact information for the national emergency centre and emergency services. Emergency services telephone numbers must be obtained from the local service providers.

5.2.5 Security Training

The Authority Holder shall ensure that all persons engaged in the transport of radioactive material receive training, including training in the elements of security awareness. Such training shall be provided or verified upon employment and periodically supplemented by retraining. Records of all security training undertaken shall be kept by the Authority Holder and shall be made available, upon request, to the employee and to the Regulatory Authority.

Security Awareness training should include the following elements:

- a) Describe the need for transport security;
- b) The nature of security-related threats;
- c) Methods to address security concerns;
- d) Actions to be undertaken if a security event occurs; and
- e) Training on the Transport Security Plan, where applicable, and the individual's role(s) in implementing the Transport Security Plan.

5.2.6 Exchange of security-related information

Authority Holders shall cooperate and with the appropriate authorities to exchange information on applying security measures and responding to security incidents.

The purpose of information sharing is to ensure uniformity of security throughout the transport; however, information sharing should only be conducted in a manner that is consistent with information protection requirements.

5.2.7 Communications

During transport, the Authority Holder should provide the capability for vehicle crew members to communicate with their company or law enforcement to request assistance. This can be executed, for example, by using mobile telephones. Communication should remain effective throughout the entire journey. Where this is not possible, predefined communication points in the journey should be agreed to in order to provide evidence that the journey is proceeding as planned without incident.

5.2.8 Security Verification

Conveyors shall perform security verification inspections of conveyances and shall ensure that these security measures remain effective during transport. This inspection should be documented using a signed and dated checklist prior to shipment departure.

Self-Inspections should be conducted to verify that security measures specified in the Prudent Management Practices and Basic Transport Security Level have been implemented as a condition for receiving a Transport Authority.

5.2.9 Receipt Verification

The Consignee should have procedures in place to verify that package contents match the expected radioactive material inventory and/or any relevant shipment documents available. The procedures should include provisions for notifying SAHPRA, Consignor, and Conveyor if radioactive material is discovered to be missing.

5.3 Enhanced Transport Security Level Requirements (Category 1 and 2)

In addition to Prudent Management Practices and Basic Transport Security Level Requirements, Enhanced Transport Security Level Requirements shall be applied to radioactive material which has been evaluated as Category 1 or 2.

5.3.1 Information Protection

Measures should be taken to protect sensitive information relating to transport operations, including detailed information on the schedule and route. Such information should be shared on a need-to-know basis, and includes the security system design and operation, response capability, detection, assessment, and delay capabilities.

In addition, computer security is critical to protecting sensitive information. Computers should be access-controlled commensurate to the level of information stored. These systems should be accessible only by persons on a need-to-know basis who have undergone a trustworthiness verification. The primary application for information protection is found in the security plan, which contains the above-mentioned details.

5.3.2 Transport Security Plan

A Transport Security Plan is required to be submitted to SAHPRA for all Category 1 and 2 transport applications.

The Transport Security Plan for category 2 sources must be reviewed annually.

If any aspect of the transport is subcontracted, the Authority Holder shall ensure that the subcontractor complies with the Transport Security Plan. **The Transport Security Plan shall include at least the following elements and shall be modified as needed to reflect changes in the threat level:**

- a) Specific allocation of responsibilities for security to competent and qualified persons with appropriate authority to carry out their responsibilities;
- b) Provision for keeping records of radioactive material packages or types of radioactive material transported;
- c) Review of current operations and assessment of vulnerability, including intermodal transfer, storage in transit, handling, and distribution as appropriate;
- d) Clear statements of transport security measures, including training and policies, response to conditions of a higher-level threat, verification of new employees and employment, operating practices, equipment, and other resources;

- e) Effective procedures and equipment for timely reporting and responding to security-related threats, in addition to breaches of security or security-related incidents;
- f) Procedures for evaluating and testing Security Plans and procedures for periodic review and update of the plans;
- g) Measures to ensure the security of sensitive transport information contained in the Security Plan;
- h) Measures to ensure that the distribution of sensitive transport information is limited on a need-to-know basis;
- i) Measures to monitor the location of the shipment; and
- j) Where appropriate, details concerning agreements on the point of transfer of responsibility for security.

For detailed instructions and an example plan and template, see Annex II of this document (NSS 9-G example TSP).

Transport Security Plans must be submitted at the time of the application for authority to transport. SAHPRA will inform the applicant on the details of how a secure submission of the application must be made.

5.3.3 Planning and Coordination

The shipper shall provide advance notification to the receiver and SAHPRA of each planned shipment. This advance notification must include details of the mode of transport and the expected shipment schedule. The Consignee shall confirm capability and readiness to accept delivery at the expected time and shall notify the shipper and SAHPRA on receipt or non-receipt within the expected delivery time frame.

In the case of routine industrial radiography camera use, or other routine Category 2 users, the details of the transport must be recorded in the source log and be made available to SAHPRA for inspection. No additional notifications are required. This does not apply in cases where a radiography camera is being shipped for the purpose of replacing the source and a notification is required.

5.3.4 Physical Security

The physical security system includes those measures that provide detection, delay, and response.

5.3.4.1 Detection

At a minimum, detection is provided by the vehicle crew who have been briefed on the Transport Security Plan and have been provided with emergency contact information. The vehicle crew should be instructed to immediately contact law enforcement as well as their RPO if any suspicious activity is observed.

NSS 9-G Section 5.48. Electronic intrusion detection and alarms, including duress alarms, should be considered. Electronic intrusion detection technologies can be suitable for providing an immediate indication of an intrusion into the cargo area. **Examples of this technology include the following:**

- (a) Balanced magnetic door switches;
- (b) Light sensors (for closed conveyances);
- (c) Fibre optic and other electronic seals;
- (d) Passive infrared, microwave or video motion detection.

If electronic intrusion detection systems are implemented, they must be remotely monitored by personnel with training on the correct procedures to follow if an alarm is detected.

Enhanced detection will be provided by armed escorts and the RPO accompanying the shipment. Each must be able to communicate with SAHPRA, SAPS, and the Emergency Centre.

5.3.4.2 Delay

Locks and seals should be applied to conveyances, compartments or freight containers. Locks and seals should be checked to confirm the integrity before dispatch, after any stops made during the journey and during any intermodal transfer of each radioactive material consignment.

If enclosed freight containers are used, verification of the integrity of a door seal is sufficient, and individual seals on packages inside the freight container do not need to be verified. Lock fittings and components, such as attachment points and tie-downs, should be complementary to the quality and strength of the locks. In cases where chains are unavailable, the package should be contained within an ISO Intermodal Container to provide an additional barrier against direct access to the package.

Procedures should be established to ensure the security of keys to conveyances and locks.

5.3.4.3 Response

Response is provided by SAPS per their operating procedures. The Authority Holder must include contact information for SAPS in the instructions provided to the crew of the vehicle.

5.3.5 Communications

Routine communications may be conducted by mobile telephone. Techniques such as code words and phrases should be used to reduce the amount of sensitive information that is shared over open communications. This does not apply in an emergency.

5.3.6 Pre-shipment Security Verification

The Consignor and Conveyor should conduct a pre-shipment security verification of the conveyance and security systems before commencing transport. The purpose of this verification is to ensure that the security measures are implemented as described in the Transport Security Plan and are functioning normally.

This verification should ensure that security measures specified in the Prudent Management Practices, Basic Transport Security Level, and Enhanced Transport Security Level have been implemented as a condition for receiving a Transport Authority.

NSS 9-G Section 5.52. Corrective actions should be taken upon identifying that one or more elements are deficient. Without corrective actions, shipments should not be undertaken.

5.4 Other Security Measures for Category 1 Shipments

In addition to Prudent Management Practices, Basic Transport Security Level Requirements, and Enhanced Transport Security Level Requirements, the following provisions shall be applied to radioactive material that has been evaluated as Category 1.

5.4.1 Advance Notification

The Consignor shall provide advance notification to the receiver of each planned shipment, mode of transport, and expected delivery time. The Consignee shall confirm capability and readiness to accept delivery at the expected time and shall notify the Consignor on receipt or non-receipt within the expected delivery time frame.

5.4.2 Shipment Monitoring and Tracking

The Authority Holder shall use tracking methods or devices to monitor the movement of packages or conveyances containing radioactive material. All packages shall be tracked during transport to ensure that the status of the package can be determined at any time. Road vehicles transporting Category 1 quantities of radioactive material shall be equipped with automatic and real-time tracking capability.

5.4.3 Additional Physical Security

Enhanced detection will be provided by armed escorts, the driver, and the radiation protection officer accompanying the shipment. Each will have a method for communication with SAHPRA, SAPS, and the National Emergency Centre.

Enhanced delay will be provided by chain tie-downs/any other tie-down that does not compromise the integrity of the transport container. In cases where chains are not available, the package should be contained within an ISO Intermodal Container to provide an additional barrier against direct access to the package.

Enhanced response will be provided by armed escorts who will be responsible for the interruption and neutralisation of a security incident according to their own operating procedures.

5.4.4 Additional Pre-Shipment Verification

Category 1 pre-shipment security verification should be performed in stages. The first stage should occur well in advance of shipments to ensure deficiencies are identified and time is available to resolve them. Final verification should be conducted using a checklist just prior to departure to ensure that all security measures included in the Transport Security Plan are in place and operational. This pre-shipment verification should be documented using a signed and dated checklist before shipment departure.

5.5 Additional Security Measures

SAHPRA may require these security measures in addition to those specified in the security provisions identified in Prudent Management Practices, Basic Transport Security Level, Enhanced Transport Security Level, and Additional Security Measures for Category 1 Shipments in view of the design basis threat, the assessment of the prevailing threat, or the nature of the material being transported.

In such cases, the SAHPRA may require the following Additional Measures to be applied to any of the above security levels on a case-by-case basis:

- 1) Additional training, beyond basic security awareness;
- 2) Formal national security clearance of transport personnel commensurate with their responsibilities;
- 3) Additional guards to accompany certain transports;
- 4) Evaluation of the potential for sabotage and associated radiological consequences for a package design;
- 5) Search of the conveyance before loading and shipment to ensure that it has not been tampered with in any way that could compromise security;
- 6) Conveyances specially designed or modified to provide additional security features such as:
 - a) Panic button
 - b) Vehicle disablement
- 7) Review of the Transport Security Plan by SAHPRA;
- 8) Exercises to ensure that contingency plans are adequately robust;
- 9) Written instructions detailing the responsibilities of personnel with specific security responsibilities;
- 10) Additional measures, consistent with national requirements, to protect the confidentiality of information; and the
- 11) Use of secure, redundant communications.

5.6 Category 1, 2, and 3 Security Guidance Table¹

Security Provision ²	Category 3	Category 2	Category 1
Graded approach applied (6.1(a))	X	X	X
Recipient is Authorised (6.1(b))	X	X	X
Security is maintained at stops (6.1(c))	X	X	X
Measures must be taken to find missing material (6.1(d))	X	X	X
Transport vehicles are lockable (6.1(e))	X	X	X
Package is approved and locked (6.1(f))	X	X	X
Package is locked in vehicle (6.1(g))	X	X	X
Package permanently fixed to vehicle (6.1(h))	X	X	X
Package should not be left unattended (6.1(i))	X	X	X
Basic package tracking (6.1(j))	X	X	X
Basic Communications (6.1(k))	X	X	X
Contingency Plans (6.2.1)	X	X	X
Trustworthiness Determination (6.2.2)	X	X	X
Carrier credentials (6.2.3)	X	X	X
Written procedures (6.2.4)	X	X	X
Basic Security Training (6.2.5)	X	X	X
Exchange of security information (6.2.6)	X	X	X
Tamper Evident Seals (6.2.7)	X	X	X
Self-Inspection/Verification (6.2.8)	X	X	X
Receipt Verification (6.2.10)	X	X	X
Protection of security related information (6.3.1)		X	X
Transport Security Plan (6.3.2)		X	X
Planning and coordination (6.3.3)		X	X
Methodical Physical Protection System (6.3.4)		X	X
Enhanced Communications (6.3.5)		X	X

¹ Additional security measures found in section 6.5 may be applied to any category of shipment as determined by SAHPRA

² Note that these security measures are described in shortened terms, refer to the section in parentheses for details of each requirement

Receipt Notification (6.3.6)		X	X
Advanced Notification (6.4.1)			X
Satellite Tracking (6.4.2)			X
Enhanced Physical Protection System (6.4.3)			X
Detailed Security Verification (6.4.4)			X

5.7 International Shipment by Air or Sea

For air transport, shipment shall be carried out in accordance with the applicable security provisions (Annexes 17 and 18 of the Convention on International Civil Aviation and the International Civil Aviation Organization Technical Instructions for the Safe Transport of Dangerous Goods by Air).

For maritime transport, shipment shall be carried out in accordance with the applicable security provisions of the International Ship and Port Facility Security Code and of the International Maritime Dangerous Goods Code as required by the International Convention for the Safety of Life at Sea (SOLAS 74 amended).

5.8 Use of International Standards and Other Publication

In lieu of specific compliance with the national legislation of South Africa, a shipment which imports radioactive material is deemed to be in compliance if it meets all the applicable requirements of IAEA guidance found in SSR-6 and NSS 9-G.

6. Inspection and Enforcement

6.1 Regulatory inspection of Vehicles and Information

The Authority Holder shall provide inspectors and other officers appointed by the Regulator with immediate access to transport vehicles and locations in which radioactive materials are present to obtain information concerning the status of security and verify compliance with the Authority conditions. Each Authority Holder shall make available to the Regulator information and records regarding security as requested.

6.2 Enforcement

An Authority Holder who violates any security-related condition of an authority, is subject to the following enforcement actions by the Regulator.

- (1) An Inspection Report requiring corrective action within a specified time;
- (2) A Notice of Violation issued by the Regulator stating the alleged violation and requiring the holder of an authority, within 20 business days of the date of the notice or other specified time, to submit a written reply, including:
 - (a) Corrective steps, if any, which have been taken, and the results achieved;
 - (b) Further corrective steps which will be taken; and
 - (c) The date when full compliance will be achieved.
- (3) An Order issued by the Regulator modifying, suspending, or revoking an authority, seizing radioactive material or placing an embargo on its use, or taking such other action as may be proper. The order:
 - (a) alleges the transgressions of the act in terms whereof the authority holder may be criminally charged;
 - (b) provides that the holder of an authority must file a written answer to the order within 20 business days of its date, or such other time as may be specified in the order;
 - (c) provides for consideration by the Regulator the answer and modification or withdrawal of the order in the discretion of the Regulator and
 - (d) states the effective date of the order.
- (4) Appeal by the Authority Holder against any decision of the Regulator may be made as follows:**
 - (a) The Authority Holder may submit a written appeal to the Regulator within 20 business days of the decision, stating the legal and/or factual grounds for reversing or modifying the decision; and
 - (b) The Authority Holder will receive a written determination by the Regulator whether to reverse or modify the decision.
- (5) Referral for criminal prosecution: the Director-General may refer wilful violations of the conditions of an Authority to the Directorate for Priority Crimes for criminal investigation.

7. Validity Clause

This guideline is valid for a period of five (5) years from the effective date of the revision or until superseded by a later revision. It will be reviewed on this timeframe or as and when required.

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8. Annex I: Categorisation Table

8.1 Table 1. Categories of Radioactive Material for Common Applications

Category	Practice	Indicative A/D
1	Irradiators (industrial, sample or blood irradiators) Fixed multibeam teletherapy (gamma knife) Teletherapy	$A/D \geq 1000$
2	Industrial gamma radiography High/medium dose rate brachytherapy	$1000 > A/D \geq 10$
3	Fixed industrial gauges that incorporate high activity radioactive sources Well logging gauges Soil density/moisture gauges	$10 > A/D > 1$

8.2 Table 2. Activities Corresponding to Category Thresholds

	Category 1		Category 2		Category 3	
Radionuclide	1000 x D		10 x D		D	
	(TBq)	(Ci) ³	(TBq)	(Ci) ²	(TBq)	(Ci) ²
Am-241	6.E+01	2.E+03	6.E-01	2.E+01	6.E-02	2.E+00
Am-241/Be	6.E+01	2.E+03	6.E-01	2.E+01	6.E-02	2.E+00
Cf-252	2.E+01	5.E+02	2.E-01	5.E+00	2.E-02	5.E-01
Cm-244	5.E+01	1.E+03	5.E-01	1.E+01	5.E-02	1.E+00
Co-60	3.E+01	8.E+02	3.E-01	8.E+00	3.E-02	8.E-01
Cs-137	1.E+02	3.E+03	1.E+00	3.E+01	1.E-01	3.E+00
Gd-153	1.E+03	3.E+04	1.E+01	3.E+02	1.E+00	3.E+01
Ir-192	8.E+01	2.E+03	8.E-01	2.E+01	8.E-02	2.E+00
Pm-147	4.E+04	1.E+06	4.E+02	1.E+04	4.E+01	1.E+03
Pu-238	6.E+01	2.E+03	6.E-01	2.E+01	6.E-02	2.E+00
Pu-239/Be	6.E+01	2.E+03	6.E-01	2.E+01	6.E-02	2.E+00
Ra-226	4.E+01	1.E+03	4.E-01	1.E+01	4.E-02	1.E+00
Se-75	2.E+02	5.E+03	2.E+00	5.E+01	2.E-01	5.E+00
Sr-90 (Y-90)	1.E+03	3.E+04	1.E+01	3.E+02	1.E+00	3.E+01
Tm-170	2.E+04	5.E+05	2.E+02	5.E+03	2.E+01	5.E+02
Yb-169	3.E+02	8.E+03	3.E+00	8.E+01	3.E-01	8.E+00
Au-198*	2.E+02	5.E+03	2.E+00	5.E+01	2.E-01	5.E+00
Cd-109*	2.E+04	5.E+05	2.E+02	5.E+03	2.E+01	5.E+02
Co-57*	7.E+02	2.E+04	7.E+00	2.E+02	7.E-01	2.E+01
Fe-55*	8.E+05	2.E+07	8.E+03	2.E+05	8.E+02	2.E+04
Ge-68*	7.E+02	2.E+04	7.E+00	2.E+02	7.E-01	2.E+01
Ni-63*	6.E+04	2.E+06	6.E+02	2.E+04	6.E+01	2.E+03
Pd-103*	9.E+04	2.E+06	9.E+02	2.E+04	9.E+01	2.E+03
Po-210*	6.E+01	2.E+03	6.E-01	2.E+01	6.E-02	2.E+00
Ru-106 (Rh-106)*	3.E+02	8.E+03	3.E+00	8.E+01	3.E-01	8.E+00
Tl-204*	2.E+04	5.E+05	2.E+02	5.E+03	2.E+01	5.E+02

The category of radioactive material shall be further adjusted based on the following factors:

³ The primary values to be used are given in TBq. Curie values are provided for practical usefulness and are rounded after conversion.

* These radionuclides are very unlikely to be used in individual radioactive sources with activity levels that would place them within Categories 1, 2 or 3.

- **Radioactive decay:** If the activity of radioactive material decays to a level below the appropriate threshold in Table 2 or below that which is normally used, the Authority Holder may re-categorise the radioactive material based on the new A/D ratio.
- **Aggregation of sources:** When multiple items containing radioactive material are in close proximity, such as in manufacturing processes (e.g., in the same room or building) or in storage facilities (e.g., in the same enclosure or container), or in a single conveyance, the Authority Holder shall aggregate the activity of the radioactive material for the purpose of categorisation. In situations of this type, the summed activity of the radionuclide shall be divided by the appropriate D value and the calculated ratio A/D compared with the ratio's A/D given in Table 2, thus allowing the set of different radionuclides to be categorised on the basis of activity. If radioactive material of various radionuclides is aggregated, then the sum of the ratio's A/D should be used in determining the category, in accordance with the formula:

$$\text{Aggregate A/D} = \sum \frac{\sum A_{i,n}}{D_n}$$

where:

$A_{i,n}$ = activity of each individual radioactive material i of radionuclide n .

D_n = D value for radionuclide n .

9. Annex II: Transport Security Plan Template Outline

BOX I–1. EXAMPLE STRUCTURE OF THE TRANSPORT SECURITY PLAN

1. **SCOPE**
2. **OBJECTIVES**
3. **DESCRIPTION OF THE SHIPMENT AND MATERIAL TO BE TRANSPORTED**
 - 1.1. Description of radioactive material
 - 1.2. Mode(s) of transport
4. **ADMINISTRATIVE REQUIREMENTS**
 - 1.1. Policies and procedures
 - 1.2. Vulnerability and threat assessment
 - 1.3. Testing and evaluating the Transport Security Plan
 - 1.4. Transport security verification
 - 1.5. Notification of relevant agencies
 - 1.6. Review and update of the Transport Security Plan
5. **PERSONNEL QUALIFICATIONS**
 - 1.1. Trustworthiness
 - 1.2. Training
6. **RESPONSIBILITIES**
 - 1.1. Organisational structure
 - 1.2. Allocation and transfer of responsibilities
7. **INFORMATION MANAGEMENT**
 - 1.1. Information security
 - 1.2. Records retention
8. **TRANSPORT SECURITY MEASURES**
 - 1.1. Routes
 - 1.2. Transport security system
 - 1.1.1. Conveyance
 - 1.1.2. Operations command and control
 - 1.1.3. Nuclear security measures
 - 1.1.4. Communications and positional tracking for normal operations

1.1.5. Maintenance and testing of systems and equipment

9. EMERGENCY RESPONSE

- 1.1. Emergency and contingency response
- 1.2. Communications during incidents
- 1.3. Reporting of threats and incidents

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BOX I-2. DETAILS TO BE CONSIDERED FOR THE TRANSPORT SECURITY PLAN

1. SCOPE

This section defines the shipments and entities that are covered in the Transport Security Plan, including the following:

- The type of radioactive material to be shipped;
- The locations of the shipper and receiver;
- The identification of the carrier;
- The regulations and requirements that were used in the development of the Transport Security Plan.

This section includes the complete legal name and address of the entity responsible for preparing and submitting the Transport Security Plan. This includes information about the shipper, carriers, receiver and other entities involved with the shipment, including guards employed for the shipment, and information about transit States when international transport is involved.

2. OBJECTIVES

This section provides a clear statement of the objectives that the plan intends to accomplish, including the following:

- Ensuring security to protect personnel, equipment, and radioactive material;
- Providing clear direction to personnel on the following actions to be taken:
 - Ensuring security of shipments;
 - Providing an appropriate response to incidents.

3. DESCRIPTION OF THE SHIPMENT AND MATERIAL TO BE TRANSPORTED

3.1. Description of radioactive material

The description of the material to be transported includes the following:

- Nature of the material;
- Type;
- Quantity (activity);
- Physical and chemical characteristics;

- Category;
- Hazards;
- Packaging;
- Number of packages in a consignment

3.2. **Mode(s) of transport**

This subsection specifies the mode(s) of transport (road, rail, air, water).

4. **ADMINISTRATIVE REQUIREMENTS**

This section states the persons, organisations and other entities involved in the transport covered by the plan. This section also provides a detailed presentation of all the administrative requirements that need to be satisfied to provide adequate security during the transport of the radioactive material.

4.1. **Policies and procedures**

This subsection lists those specific policies and procedures issued either by State entities or the responsible party that apply to the shipment(s). These policies and procedures are specified in the following:

- Policies and operational procedures for consistent implementation of security measures addressed in the Transport Security Plan;
- Contingency plans for responding to malicious acts during transport, recovery of lost or stolen material and mitigation of consequences.

4.2. **Vulnerability and threat assessment**

As applicable, this subsection describes how the shipper and carrier will ensure that security measures are adequate by performing a vulnerability assessment that considers the threat level.

The vulnerability assessment includes a review of planned operations (equipment operability) and identification of potential vulnerabilities. This includes evaluating shipment-specific parameters such as modes of transport, intermodal transfers, overnight stops and information protection.

The threat level used is described and a description of how threat level changes will be communicated and acted upon is included. Changes in circumstances that might necessitate evaluating the need for operational changes are identified in this section, such as activities that might impact routing (e.g.,

activists or demonstrations, road conditions, traffic conditions, secure parking for overnight trips).

4.3. Testing and evaluating the Transport Security Plan

This subsection specifies the procedures for evaluating and testing the effectiveness of the Transport Security Plan.

4.4. Transport security verification

This subsection describes how the shipper and carrier will ensure that all specified security measures are in place and operational prior to initiating a shipment. Any planned use of checklists for performing pre-shipment security verification and any corrective actions are outlined here.

4.5. Notification of relevant agencies

This subsection specifies responsibility for the timing and method of communicating notifications to relevant agencies (before, during, and after transportation).

4.6. Review and update of the Transport Security Plan

This subsection specifies when and how the reviews and updates of the Transport Security Plan are to be accomplished.

5. PERSONNEL QUALIFICATIONS

5.1. Trustworthiness

This subsection describes the level of trustworthiness needed for personnel involved in transport. It describes the process used to verify trustworthiness at each of those levels.

5.2. Training

This subsection specifies training qualifications for personnel involved in transport, including the nature and frequency of the training. It also includes a description of any exercises that will be conducted as well as the schedule that will be followed for each type of exercise. A description is also included of how the results of exercises will be evaluated, including documentation of the results of the exercises and any corrective actions taken.

6. RESPONSIBILITIES

This section specifies how responsibilities are assigned and how they are transferred as shipments proceed.

6.1. Organisational structure

This subsection specifies the organisational structure of the entities involved in the transport, describing the chain of command, including the names of responsible personnel.

6.2. Allocation and transfer of responsibilities

This subsection describes the responsibilities of all organisations and persons engaged in the transport of radioactive material, including how and when security responsibilities are transferred.

7. INFORMATION MANAGEMENT

This section specifies how all information will be managed, particularly for security-sensitive information. Reference to other information management procedures can be used.

7.1. Information security

This subsection describes how the security of information will be ensured. This description might include identification of sensitive information, classification review and marking, reproduction restrictions, distribution (i.e., authorised access and need-to-know), storage requirements and destruction.

7.2. Records retention

This subsection identifies who has responsibility for retaining records to ensure records are handled in accordance with regulatory requirements and procedures (can include requirements for shippers, carriers and receivers).

8. TRANSPORT SECURITY MEASURES

This section describes the specific security measures that have been established for shipment, addressing those measures that apply before transport, during transport (including storage incidental to transport), and upon receipt of the radioactive material.

8.1. Routing

This subsection specifies routes and associated in-transit storage and intermodal transfer locations.

The information provided here includes the following:

- Planned (primary) and alternate routes for all modes of transport, including criteria for when the alternate routes will be used;
- Process for pre-shipment evaluation of routes, assessment of vulnerabilities;
- Identification of any in-transit storage and intermodal transfers, including security

arrangements.

8.2. Transport security system

This subsection describes the security system, including specific security measures (based on the transport security level of the shipment) and other arrangements that will be used.

8.2.1. Conveyance

This subsection describes the conveyances (road, rail, air, water), including any special requirements for the conveyances.

8.2.2. Operations command and control

This subsection identifies command and control procedures for normal and emergency operations. This information includes a chain of command structure, decision-making authority, points of contact and identification of response agencies.

Nuclear security measures

This subsection identifies the nuclear security measures to be used during transport. These measures include those used to provide detection, delay and response. Examples of these measures include the following:

- Tamper-indicating devices and seals (packages and conveyances);
- Locks (single or multiple) for packages, cargo compartment, and conveyance (e.g., door keys, ignition keys);
- Secure tie-downs and over-packs;
- Immobilising devices.

This subsection also identifies the process for authorising alternative measures (i.e., when a feature is not operational or available).

8.2.3. Communication and positional tracking for normal operations

This subsection describes the structure of the primary and alternative communication systems for the transport operation. It also describes any system for tracking the conveyances, including identification of the location at which shipment monitoring will occur.

8.2.4. Maintenance and testing of systems and equipment

This subsection addresses how all of the systems involved in the shipment(s) (such as

communications and tracking) are maintained and tested. This subsection also addresses the checking and testing of all mission-related equipment that will be performed before the beginning of the transport. It also specifies periodic testing requirements.

BOX I-2. DETAILS TO BE CONSIDERED FOR THE TRANSPORT SECURITY PLAN (cont.)

1. EMERGENCY RESPONSE

Emergency response includes both tactical and non-tactical (i.e., non-security-related emergency) planning. This section identifies the range of incidents that might need response measures to be initiated, describes the appropriate response measures, and clearly defines the response resources.

1.1. Emergency and contingency response

This subsection identifies handling responses of non-nuclear or non-radiological emergency and security-related incidents; it is not to be confused with specific arrangements for response to a nuclear or radiological emergency. Response actions to be included in this subsection are actions that will be taken by crew members, the transport control centre or other operations centres; shipper and receiver technical support staff; emergency response units along the route; escort personnel (if present); guard or security force (if present), and response forces.

Emergency situations can include road closure, vehicle breakdown, vehicle accidents, and driver illness. Corresponding emergency arrangements could include the availability of backup vehicles and drivers, capabilities for towing and lifting, and plans for use of safe havens.

This subsection describes any necessary advance information to respond to forces along the route, including the time within which the communication of information must be completed before the shipment. It also identifies any accompanying guard or security force.

1.2. Communications during incidents

This subsection includes a description of the communications systems and actions that will be taken to address both emergency and nuclear security events. This information could include the types of communications equipment used and features to ensure the security of communications.

1.3. Reporting of threats and incidents

This subsection describes reporting requirements, including types of events that must be reported, to whom and how the event will be reported, and the timeframe for reporting.